

SPACE TRAVEL REQUEST & MISSION MANAGEMENT PLATFORM

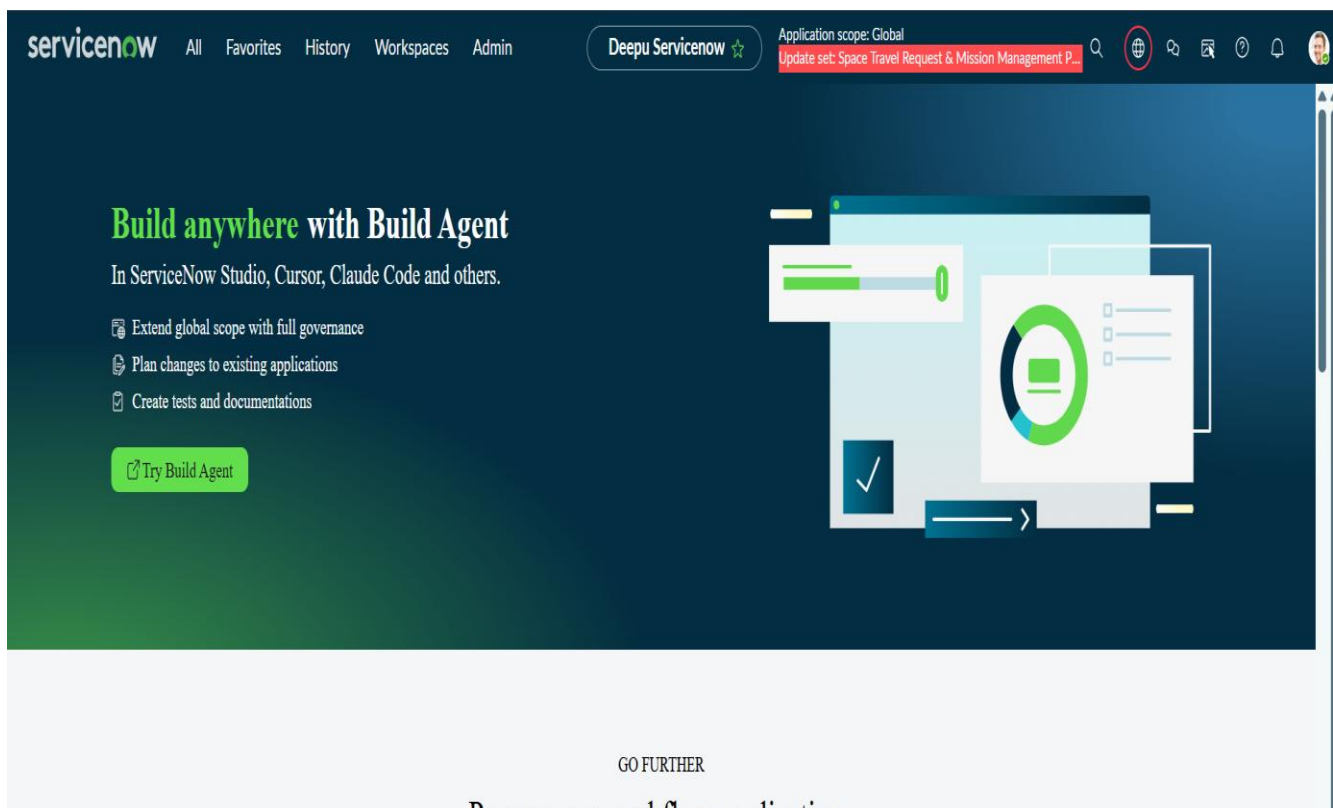
ServiceNow Project Documentation

Technology: ServiceNow

Platform: App Engine Studio / Workflow Studio

Project Type: ServiceNow Application Development

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[Application Home Page]

Figure 1: Application Home Page.

2. Abstract

The Space Travel Request & Mission Management Platform is a custom ServiceNow application designed to manage the lifecycle of space travel requests and missions. The application maintains traveler, destination, spacecraft, request, mission, crew, and resource information in a centralized platform.

The solution uses App Engine Studio for application and table development and Workflow Studio/Flow Designer for process automation. A submitted travel request enters an approval process. When the request is approved, the workflow automatically creates a mission; when it is rejected, the request follows the rejection path. Reports and a dashboard provide visibility into requests, missions, priorities, destinations, and spacecraft status.

3. Introduction

3.1 Background

Space travel operations involve several connected entities, including travelers, destinations, spacecraft, approvals, missions, crew members, and resources. Managing these activities manually can make tracking and coordination difficult. ServiceNow provides a platform for creating custom applications and automating business processes.

3.2 Problem Statement

A centralized system is required to manage space travel requests and convert approved requests into organized missions while maintaining related operational information.

3.3 Objectives

- Create a centralized space travel management application.
- Maintain structured information using custom tables.
- Automate travel request approval.
- Automatically create missions after approval.
- Manage mission crew and resources.
- Provide notification capability.
- Provide reports and dashboard-based monitoring.

3.4 Scope

The project covers traveler management, destination management, spacecraft management, travel requests, approvals, mission management, crew management, resources, notifications, reports, and dashboards. AI functionality is not included in the current implementation.

4. Existing System and Proposed System

4.1 Existing Problems

- Scattered information
- Manual approval tracking

- Repeated data entry
- Difficulty tracking missions
- Limited visibility into request status
- Limited centralized reporting

4.2 Proposed ServiceNow Solution

The proposed solution uses ServiceNow to create a centralized application in which travel requests are connected to travelers, destinations and spacecraft. The approval workflow controls the transition from request submission to approval or rejection, and an approved request can automatically result in a mission record.

4.3 Advantages

- Centralized data
- Automated workflow
- Reduced manual processing
- Better request tracking
- Automatic mission creation
- Structured mission management
- Reports and dashboard visibility

5. Technologies Used

Technology / Component	Purpose
ServiceNow	Application platform
Application Engine Studio	Application and custom table development
Custom Tables	Store project-specific data
Workflow Studio / Flow Designer	Workflow automation
Approval Action	Request approval
Notifications	Notification configuration
Reports	Data analysis and monitoring
Dashboard	Centralized visualization

6. System Architecture

6.1 Overall Architecture

The application is organized around master data, travel requests, and missions. Space Traveler, Spacecraft, and Destination provide supporting data for Space Travel Request. An approved request leads to Mission, which is then related to Mission Crew and Mission Resource.

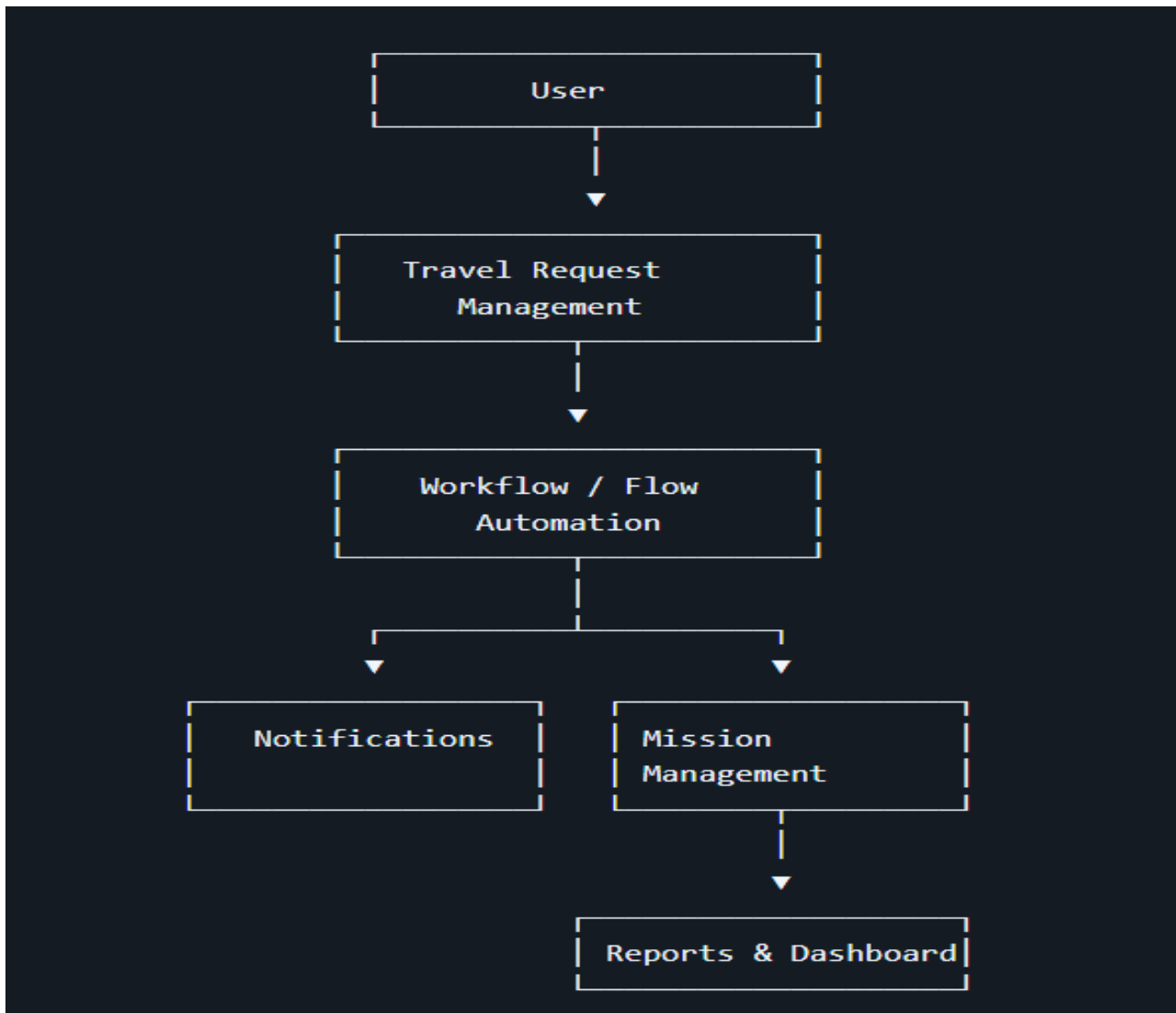
Space Traveler → Space Travel Request → Mission → Mission Crew / Mission Resource

6.2 Workflow Diagram

Travel Request Created/Updated → Status Submitted → Under Review → Ask For Approval → Approved or Rejected. If approved, the request is updated to Approved and a Mission record is created. If rejected, the request follows the rejection path.

6.3 Data/Table Relationships

The project uses relationships between the custom tables so that a request can reference its traveler, destination, and spacecraft, while a mission can reference the originating travel request and its operational crew and resources.



7. Application Modules

7.1 Space Traveler

Stores traveler information such as name, email, phone, nationality, traveler type, space experience, emergency contact, and status.

Space Traveler

SPA0001010

Update

Delete

↑

↓

Number

SPA0001010

Created

2026-09-04 23:16:40

Full Name

Deepu

Email

deepu@gmail.com

DOB

2019-05-15

Nationality

indian

Phone

India

Created by

admin

+91 91258 63485

Emergency Contact Name

jyothi

Emergency Contact Phone

India

Traveler Type

Tourist

Space Experience

Active

+91 93214 78563

Update

Delete

[\[SN Utils\] Versions \(0\)](#)

Figure 5: Space Traveler Form.

7.2 Destination

Stores available travel destinations and related information such as destination type, distance, travel time, risk level, and status.

Destination

DES0001002

Update

Delete

↑

↓

Number

DES0001002

Destination Name

Mars

Destination Type

Planet

Distance

225,000,000

Travel Time

7 Months

Risk Level

High

Status

Available

Update

Delete

[\[SN Utils\] Versions \(0\)](#)

Figure 6: Destination Form.

7.3 Spacecraft

Stores spacecraft information including spacecraft name, type, capacity, fuel capacity, current location, maximum range, status, and maintenance information.

Spacecraft

CRA0001002

Update

Delete

↑

↓

Number

CRA0001002

Current Location

Earth

Spacecraft Name

Star One

Spacecraft Type

Passenger Shuttle

Capacity

10

Maximum Range

100,000

Status

Available

Fuel Capacity

50,000

Last Maintenance Date

2026-06-01

Next Maintenance Date

2026-12-01

Update

Delete

[SN Utils] Versions (0)

Figure 7: Spacecraft Form.

7.4 Space Travel Request

This is the central request table. It contains Traveler, Destination, Spacecraft, Travel Date, Number of Travelers, Purpose, Priority, Status, Special Request, Approval Comments, and Approver. The request status choices used in the application include Draft, Submitted, Under Review, Approved, Rejected, Completed, and Cancelled.

Space Travel Request

RE0001011

Update

Delete

↑

↓

Number

RE0001011

* Traveler

SPA0001010

* Destination

DES0001002

Spacecraft

CRA0001002

* Travel Date

2026-11-17

* Number of Travelers

1

* Purpose

Tourism

Priority

Medium

Status

Approved

Special Request

window seat

Approval Comments

Approver

Update

Delete

[SN Utils] Versions (0)

Figure 8: Space Travel Request Form.

7.5 Mission

Stores missions associated with approved travel requests. Fields include Mission Name, Travel Request, Destination, Spacecraft, Launch Date, Mission Status, Mission Type, and Notes. Mission Status choices include Planned, Ready, Launched, In Progress, Completed, and Cancelled.

Mission Resource
 RES0001002

Number: RES0001002
 Resource Name: Oxygen Supply
 Quantity: 500
 Status: Available

Mission: MIS0001002
 Resource Type: Oxygen
 Unit: Liters
 Notes: Oxygen supply for Mars mission

Update Delete

[\[SN Utils\] Versions \(0\)](#)

Version	Created By	Created Date	Updated By	Updated Date
1.0.0				

Figure 11: Mission Resource Record.

8. Workflow Automation

The main automation was implemented using Workflow Studio / Flow Designer.

The flow is configured for the Space Travel Request Created or Updated trigger with the condition that Status is Submitted.

The flow first updates the request to Under Review and then uses the Ask For Approval action. The configured approver used during testing was Abel Tutor. The flow evaluates the approval result.

If the approval state is Approved, the request is updated to Approved and a Mission Record is created. The mission is populated from the travel request with information such as mission name, traveler, destination, spacecraft, travel date, and related values configured in the action.

If the approval is not approved, the Else branch updates the Space Travel Request according to the configured rejection path.

Workflow Studio Space Travel Request Approval Flow • Space Travel Request Approval Flow

Space Travel Request Approval Flow Active

View: [Icons] [Test] [Edit flow] [Deactivate] [More]

1 Update Space Travel Request Record

2 Ask For Approval on Space Travel Request

Action Properties

Action: Ask For Approval

Action Inputs

* Record: Trigger - Space Travel Request

Table: Space Travel Request [x_189456...]

Approval Reason: Space travel request requires mission manager approval.

Approval Field: Approver

Journal Field: Select a field

* Rules: Approve or Re... When: Anyone approves or rejects

Add another OR rule set

Data Collapse All

- Flow Variables
- Trigger - Record Created or Updated
 - Space Travel Request Rec... Record
 - Changed Fields Array.Object
 - Space Travel Request Table Table
 - Run Start Time UTC Date/Time
 - Run Start Date/Time Date/Time
- 1 - Update Record
 - Space Travel Request Rec... Record
 - Space Travel Request Table Table
- Action Status Object
- 2 - Ask For Approval
 - Approval State Choice
 - Action Status Object

Space Travel Request Approval Flow Active

View: [Icons] [Test] [Edit flow] [Deactivate] [More]

2 Ask For Approval on Space Travel Request

3 If 2 - Ask For Approval on Space Travel Request Approval State is Approved

4 then Update Space Travel Request Record

Action Properties

Action: Update Record

Action Inputs

* Record: Trigger - Space Travel Request

* Table: Space Travel Request [x_189456...]

* Fields: Status Approved

+ Add field value

Close

Data Collapse All

- Flow Variables
- Trigger - Record Created or Updated
 - Space Travel Request Rec... Record
 - Changed Fields Array.Object
 - Space Travel Request Table Table
 - Run Start Time UTC Date/Time
 - Run Start Date/Time Date/Time
- 1 - Update Record
 - Space Travel Request Rec... Record
 - Space Travel Request Table Table
- Action Status Object
- 2 - Ask For Approval
 - Approval State Choice
 - Action Status Object

Space Travel Request Approval Flow Active

View: [Icons] [Test] [Edit flow] [Deactivate] [More]

5 Create Mission Record

Action Properties

Action: Create Record

Action Inputs

* Table: Mission [x_1894564_space_0_mi...]

Mission Name: Mars

Travel Request: Trigger - Record ...

Destination: Trigger - Record ...

Spacecraft: Trigger - Record ...

* Fields: Launch Date Planned

Mission Status: Tourism

Notes: Automatically created after travel request

Data Collapse All

- Flow Variables
- Trigger - Record Created or Updated
 - Space Travel Request Rec... Record
 - Changed Fields Array.Object
 - Space Travel Request Table Table
 - Run Start Time UTC Date/Time
 - Run Start Date/Time Date/Time
- 1 - Update Record
 - Space Travel Request Rec... Record
 - Space Travel Request Table Table
- Action Status Object
- 2 - Ask For Approval
 - Approval State Choice
 - Action Status Object

Space Travel Request Approval Flow
Active
View: [Grid Icon] [List Icon]

6 Else

7

Update Space Travel Request Record

Action Properties

Action

Update Record

Action Inputs

* Record

Trigge... ▶ Space Travel Reque... ✕

* Table

Space Travel Request [x_189456... ▼]

Status ▼

Rejected ▼

* Fields

+ Add field value

Close

Figure 12: Space Travel Request Approval Flow
and
Create Mission Record Action Configuration.

*Rejection of Approval:

Approvals				
Created	Search			
Rejected approval for Space Travel Request: RE0001007				
All				
State	Approver	Comments	Approval for	Created
Rejected	Abel Tuter	2026-09-03 22:54:27 - System Administrat	(empty)	2026-09-03 22:50:57
Approved	Abel Tuter		(empty)	2026-09-02 22:24:44
Approved	Abel Tuter		(empty)	2026-09-02 05:07:08
Approved	Abel Tuter		(empty)	2026-09-02 03:40:55
Approved	Abel Tuter		(empty)	2026-09-02 03:27:41
Rejected	Fred Luddy		(empty)	2026-06-11 12:19:33
Requested	Fred Luddy		(empty)	2026-06-11 12:17:03
Requested	Fred Luddy		(empty)	2026-06-11 12:15:44
Approved	Eric Schroeder		CHG0000009	2026-06-10 23:31:51
Approved	Fred Luddy		CHG0000009	2026-06-10 23:31:51
Approved	Bow Ruggeri		CHG0000009	2026-06-10 23:31:50
Approved	David Loo		CHG0000009	2026-06-10 23:31:50
Requested	cab approver		CHG0000079	2024-08-30 06:35:36
Requested	Bernard Laboy		CHG0000079	2024-08-30 06:35:36
Requested	Luke Wilson		CHG0000079	2024-08-30 06:35:35
Requested	Howard Johnson		CHG0000079	2024-08-30 06:35:35
Requested	Christen Mitchell		CHG0000079	2024-08-30 06:35:35

servicenow All Favorites History Admin : Deepu Servicenow - RE0001007 | Space Travel Reque... Application scope: Space Travel Req... Update set: Default 1 [Space Travel ...

Space Travel Request RE0001007

Number RE0001007

* Destination DES0001002

* Travel Date 2026-10-15

* Purpose Tourism

Status Rejected

Approval Comments

* Traveler SPA0001006

Spacecraft CRA0001002

* Number of Travelers 1

Priority Medium

Special Request

Approver

Update Delete

[SN Utils] Versions (0)

9. Notifications

An approval notification was configured for the Space Travel Request process. The intended behavior is to notify the relevant recipient when the request reaches the configured approval state.

During development, the notification configuration was created and tested. However, final email delivery was not consistently verified in the current implementation. Therefore, email delivery should be treated as a pending final-test item rather than a confirmed successful result.

servicenow All Favorites History Admin : Deepu Servicenow - Space Travel Request Approved |... Application scope: Space Travel Request & Mission Ma... Update set: Default [Space Travel Request & Mission ...

Notification [sys_event_email_action] Space Travel Request Approved

Use Notifications to notify users about specific events. When configuring a notification, you can specify:

- When to send the notification
- Who receives the notification
- What content is in the notification

Name Space Travel Request Approved

* Table Space Travel Request

Mandatory

* Category Unclassified

A digest combines and summarises multiple updates into a single email.

Allow Digest

When to send Who will receive What it will contain

Notifications can be sent (if the specified Condition is met):

- A record is Inserted or Updated into the table
- The specified event is fired
- Via a Flow Action

Notification Preview

Preview records for this breakdown source

Event Creator System Administrator Preview Record RE0001003

Users

Subject Space Travel Request Approved

Body

Hello,

Your space travel request has been approved.

Request Number: RE0001003
Destination: DES0001002
Travel Date: 2026-12-15
Status: Approved

Thank you,
Space Travel Management Team

[Unsubscribe](#) | [Notification Preferences](#)

Figure 13: Notification Configuration.

10. Reports and Dashboard

Reports were created to provide operational visibility into travel requests, missions, and spacecraft.

Report	Purpose
Travel Requests by Status	Shows the distribution of requests across their lifecycle statuses.
Travel Requests by Destination	Shows request volume by destination.
Travel Requests by Priority	Shows the distribution of request priorities.
Missions by Status	Shows the current status distribution of missions.
Spacecraft by Status	Shows spacecraft according to their configured status.
Space Travel Requests Over Time	Provides a time-based view of requests; this report requires final data/configuration verification.

10.1 Space Travel Management Dashboard

The Space Travel Management Dashboard is intended to provide a centralized view of the reports created for the application. It brings request, destination, priority, mission, and spacecraft information together for easier monitoring.

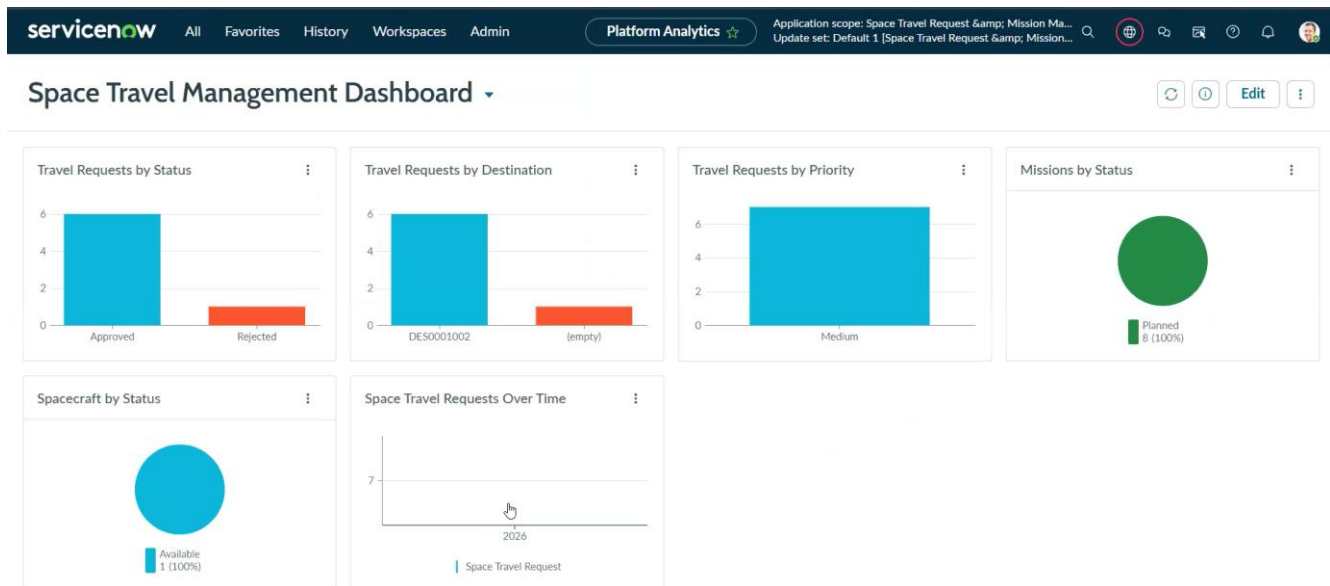


Figure 16: Space Travel Management Dashboard.

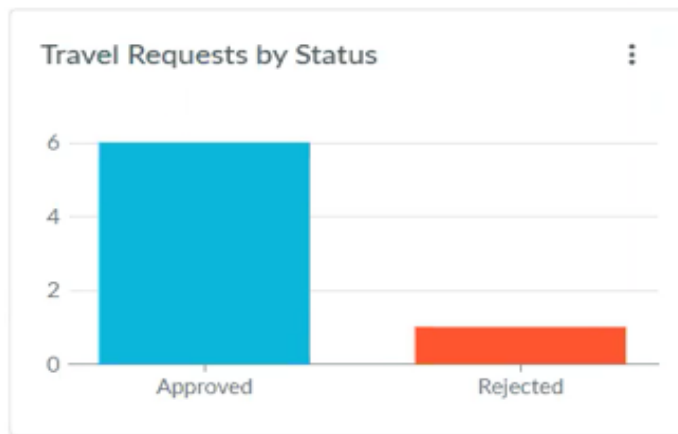


Figure 17: Travel Requests by Status Report.

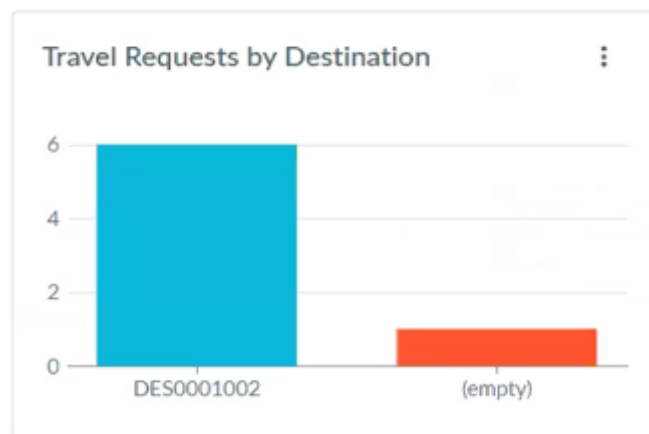


Figure 18: Travel Requests by Destination Report.

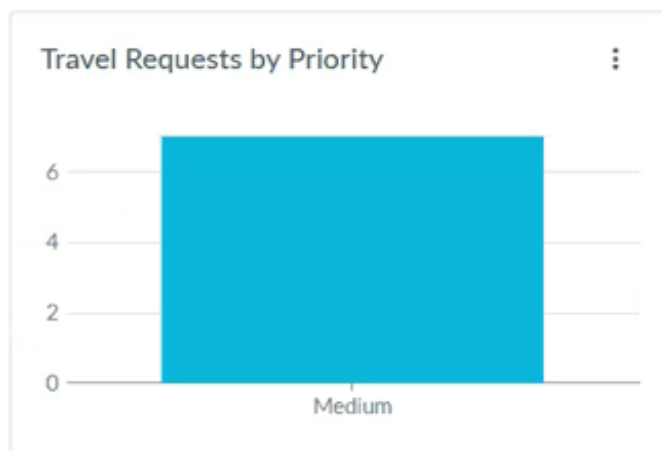


Figure 19: Travel Requests by Priority Report.



Figure 20: Missions by Status Report.



Figure 21: Spacecraft by Status Report.

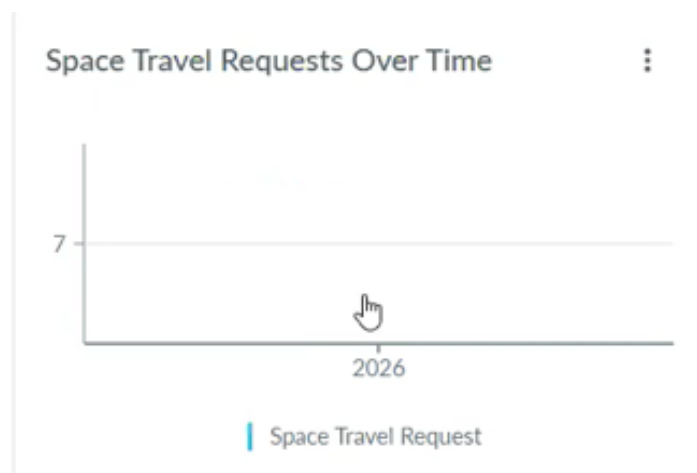


Figure 22: Space Travel Requests Over Time Report.

11. Testing

Final testing should validate the complete business process. The table below is a test checklist. PASS/FAIL should be entered only after the corresponding scenario is actually verified.

Test Case	Expected Result	Status
Record creation	Traveler, spacecraft, destination, request, mission, crew and resource records can be created.	Pass
Request submission	A request can be submitted and enters the configured workflow.	Pass
Approval	The configured approver can approve the request and the request becomes Approved.	Pass
Rejection	A rejected request follows the rejection path and does not create an approved mission.	Pass
Automatic mission creation	An approved request creates the configured Mission record.	Pass
Notification	The configured notification is generated and delivered to the intended recipient.	Pass
Reports	Reports display the expected records and categories.	Pass
Dashboard	Dashboard components display the configured reports and current data.	Pass

12. Results

The project established a ServiceNow-based application containing the planned custom tables, related records, request approval workflow, automatic mission creation logic, mission crew and resource management, notification configuration, reports, and dashboard.

During development, the approval flow was executed successfully and an approved request resulted in a Mission record with Planned status. Notification delivery and some reporting behavior require final verification and should not be presented as confirmed results until tested again.

13. Limitations and Future Scope

13.1 Current Limitations

- AI functionality was not implemented.
- Real spacecraft telemetry is not connected.
- External space/travel APIs are not integrated.
- Email notification delivery requires final verification.
- The time-based request report requires additional verification.

13.2 Future Enhancements

- AI assistance for request summaries, classification, and recommendations.
- Predictive analytics for travel demand, mission delays, and resource requirements.
- A mobile application for travelers and mission managers.
- Real-time spacecraft data and telemetry integration.
- External API integration for destination, spacecraft, scheduling, or related data.

14. Conclusion

The Space Travel Request & Mission Management Platform demonstrates how ServiceNow can be used to build a customized operational application using structured data, workflow automation, approvals, mission management, notifications, reporting, and dashboards.

The central business process is Travel Request → Approval → Automatic Mission Creation. The current version focuses on ServiceNow workflow and application management rather than AI. The application provides a foundation that can be extended with intelligent features, mobile support, real-time data, and external integrations.

15. References

- ServiceNow Platform Documentation
- ServiceNow Application Engine Studio documentation
- ServiceNow Flow Designer / Workflow Studio documentation
- ServiceNow Reporting and Dashboard documentation

16. Appendix

A. Main Application Flow

Space Traveler → Space Travel Request → Approval → Mission → Mission Crew / Mission Resource

Mission Resource	x_1894564_space_0_mission_resource
Mission Crew	x_1894564_space_0_mission_crew
Mission	x_1894564_space_0_mission
Space Travel Request	x_1894564_space_0_space_travel_request
Destination	x_1894564_space_0_destination
Spacecraft	x_1894564_space_0_spacecraft
Space Traveler	x_1894564_space_0_space_traveler

Figure 24: Application Tables.

B. Suggested Demonstration Order

Order	Demonstration Area
1	Problem and project objective
2	Application home
3	Traveler, spacecraft and destination
4	Create/submit a Space Travel Request
5	Approval workflow
6	Approved request and automatic Mission creation
7	Mission Crew and Mission Resource
8	Notification configuration and verification
9	Reports
10	Space Travel Management Dashboard
11	Rejection scenario and final conclusion

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Role: ServiceNow Developer

Platform: ServiceNow Personal Developer Instance (PDI)

Technologies Used:
ServiceNow App Engine Studio • Flow Designer • Notifications • Reports & Dashboards

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